Campus Connect Sampler

Integrated Automation



AUTOMATION INDUSTRY ASSOCIATION

Overview of Automation

Introduction to Automation

This course is intended to sharpen your creative mind with the essential understanding of industry relevant 'automation **ware.**' By the end of this course, you will confidently relate with many industrial applications and begin to ideate about different ways to monitor and control:

- Productivity
- Quality
- Wastage
- Safety

Beyond Classroom BC1

Is every machine a robot? If not, what differentiates a robot from a machine? Can you quote some examples of machines which are not robots?

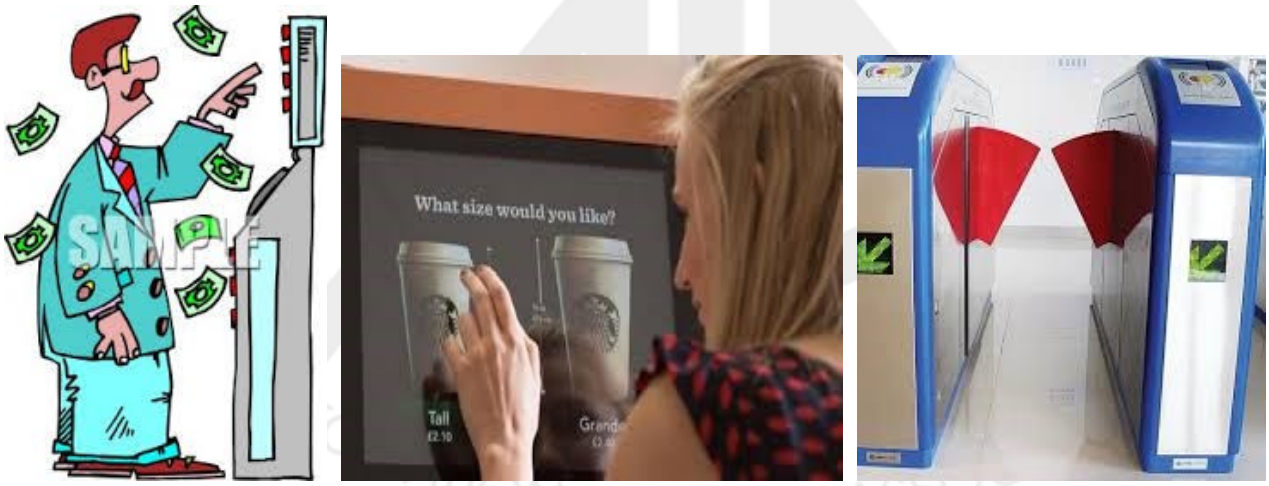


Fig: Automation in day to day life

Do the above images ring a bell?

You walk into an ATM and walk out with cash.

You press a button at Starbucks and drink the coffee poured out of the wending machine.

You swipe your token and hop onto the next metro that comes along.

Are these all examples of automation?

If so, what is it that defines automation in any process or service?

The dictionary defines "automation "as "the techniques of making an apparatus, a process or a system operate with minimal intervention." An automated system is designed to accomplish some useful action. The actions performed by automated systems are generally of two types: (1) processing and (2) transfer and positioning.

In the first case, energy is applied to accomplish some processing operation on some entity. The process may involve the shaping of metal, the molding of plastic, the switching of electrical signals in a communication system, or the processing of data in a computerized information system.

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All these actions entail the use of energy to transform the entity (e.g., the metal, plastic, electrical signals, or data) from one state or condition into another more valuable state or condition. The second type of action—transfer and positioning—is most readily seen in automated manufacturing systems designed to perform work on a product. In these cases, the product must generally be moved (transferred) from one location to another during the series of processing steps.

At each processing location, accurate positioning of the product is generally required. In automated communications and information systems, the terms transfer and positioning refer to the movement of data (or electrical signals) among various processing units and the delivery of information to output terminals (printers, video display units etc.) for interpretation and use by humans.

Beyond Classroom BC2

The operation of elevators is an example of automation in everyday life. What are your everyday experiences with automation?

Automation is a step beyond mechanization. While mechanization provides human operators with machinery to assist them augmenting the muscular requirements of work; automation greatly reduces the need for human intervention via their sensory and mental/intelligence faculties.

Beyond Classroom BC3

A pedal-operated sewing machine at your next-door tailor shop is an example of mechanization. Can you quote for more such examples?

In technical terms **"automation"** is defined as the **"delegation of action and control sequences to the associated apparatus, process or a system, by applying appropriate sensing, computational, communication and actuation technologies."**

Technology and equipment have brought a sea change in manufacturing, process and industrial environment especially since the advancement of microprocessors and the emergence of mechatronics.

In the manufacturing industry, engineers are responsible for **"applying their imagination and technical understanding of automation, to generate solutions to control and monitor material movement, its conversion, assembly and finishing into saleable products**

Where do we find Automation?

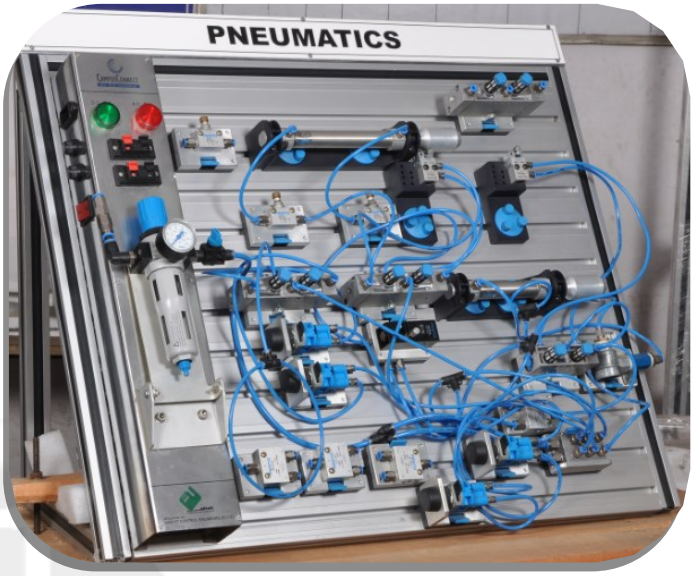
- Cars
- Automated Parking System
- Home Automation
- Facility lighting
- Petrol Dispensing
- Amusement parks

Click Here to read more……

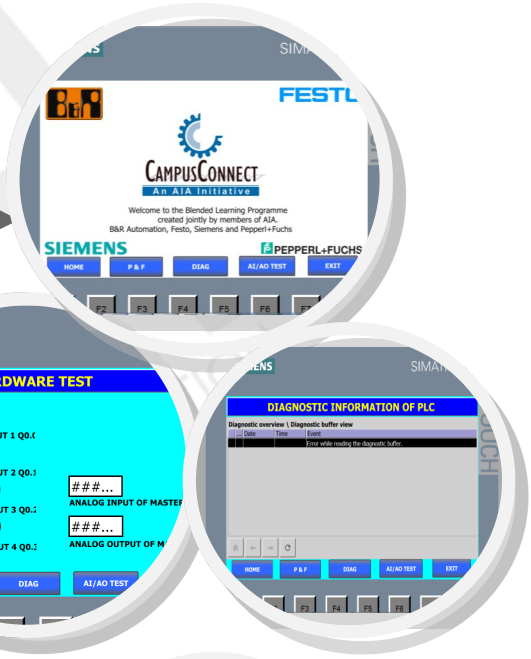
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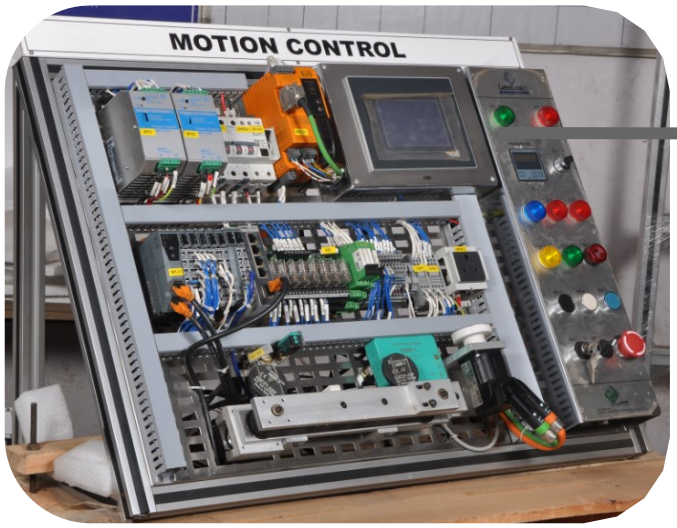
Lab Kits

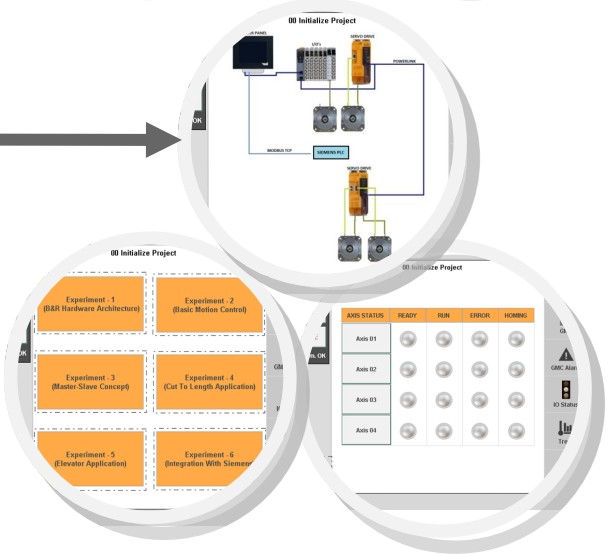












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Start up Exercises

Exercise on Sensors

Connect and operate following types of photoelectric sensor and observe their sensing behavior with different targets and change in their outputs.

- Through Beam Photo Sensor
- Retro reflective Photo Sensor with Reflector
- Distance Measurement Sensor

Exercise on Control & Visualisation

Design a ladder diagram for a conventional hardware relay logic control such that the circuit will perform each of the logic follows simultaneously when a NC push button is pressed.

- Switch a pilot light on
- De energized a solenoid coil which is previously energized.
- Start a motor
- Sound a horn.

Exercise on Motion Control

Write a program for Gearing and Synchronization (Master/Slave Concept) of 2 axes**.**

- Master Axes : Acopos Micro servo double axes channel one.
- Slave Axes : Acopos Micro servo double axes second channel.

Exercise on Pneumatics :

A double acting cylinder has to extend at a slow speed when a push botton is pressed and reach a position 75 mm from the home position and wait there for a 15 second process delay and only then the cylinder has to return back to home position at its nominal speed.

Exercise on Electro pneumatics :

Design a circuit to control speed of the double acting cylinder with two independent flow control valve to adjust the speed of forward and return stroke of the cylinder. The cylinder has to advance when the electric limit switch along with the signal from the push button is available. The return stroke is initiated when the forward end electric limit switch is triggered.

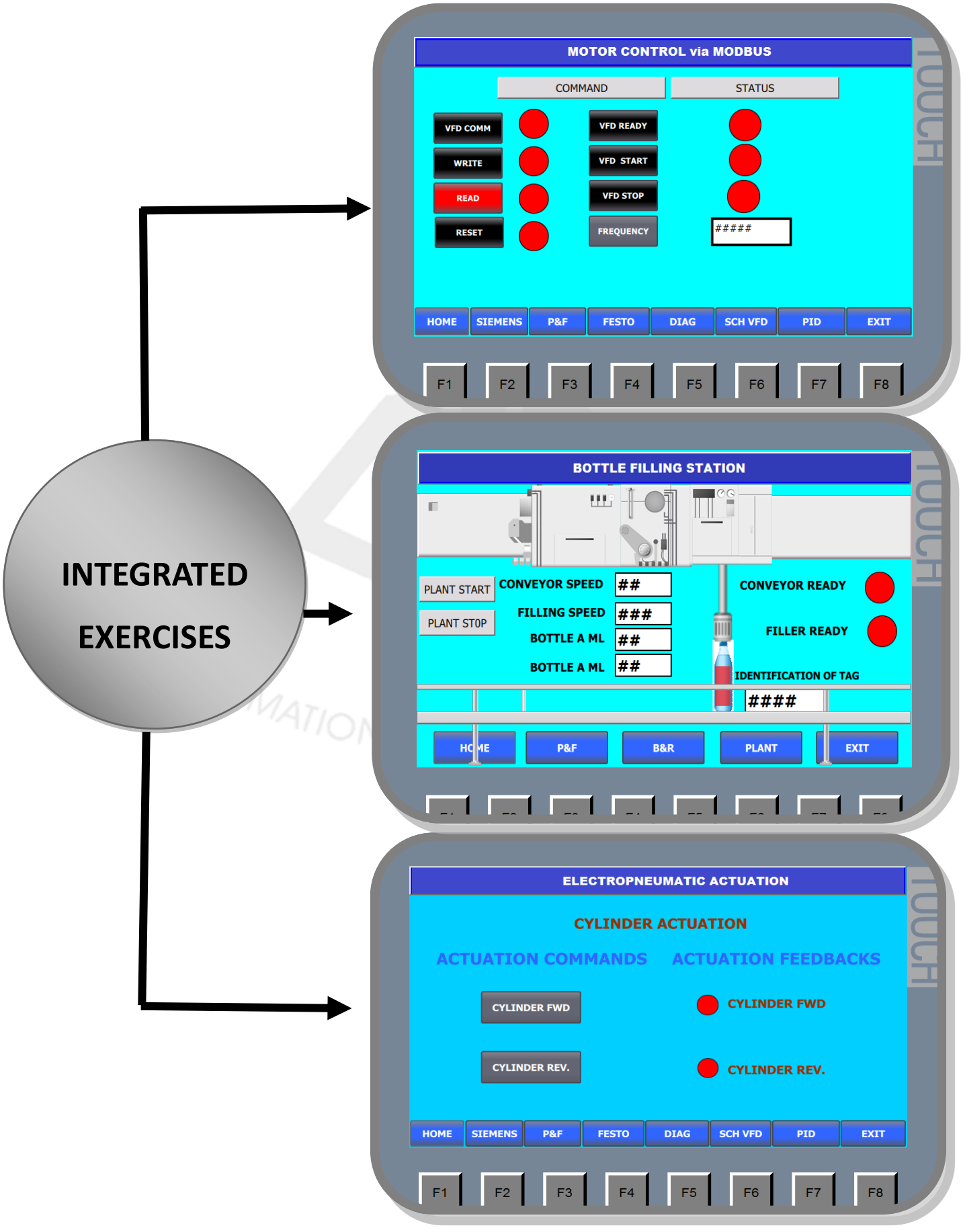
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Integration Stands



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Integrated Exercises



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EDUCATION FOR AUTOMATION



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